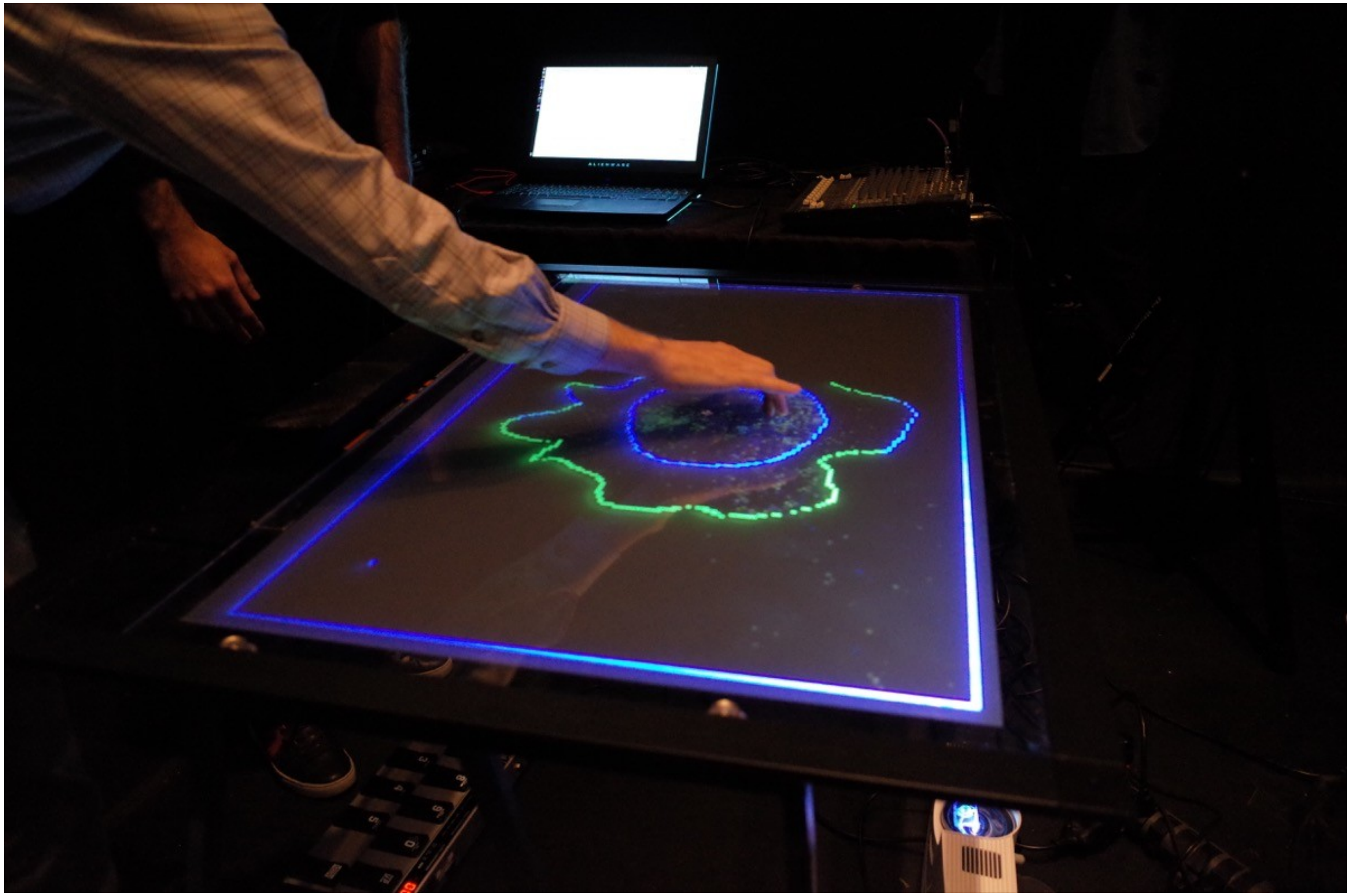

SYDE252 - W06B – System Problems

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Problem 1

A real LTIC system with input x(t) and output y(t) is described by:

$$(D^3 + 9D)\{y(t)\} = (2D^3 + 1)\{x(t)\}$$

- What is the characteristic equation of this system?
- What are the characteristic modes?
- Assuming:

$$y_{zir}(0) = 4, \quad \dot{y}_{zir}(0) = -18, \quad \ddot{y}_{zir}(0) = 0$$

... find the system's ZIR. Simplify so the ZIR is not complex.

Problem 2

The unit impulse response of an LTIC system is:

$$h(t) = (2e^{-3t} - e^{-2t})u(t)$$

... find the system's ZSR for the following inputs:

$$x(t) = u(t)$$

$$x(t) = e^{-t}u(t)$$

Problem 3

Sketch an Impulse response, h(t), of a non-causal LP system that has an approximate cutoff frequency of 5 kHz. There are many possible solutions, so explain the important features of your h(t).

FYI:

$$T_h = T_r$$

$$T_h = \frac{\int_{-\infty}^{\infty} h(t)dt}{h(t_0)}$$

For an idealized pulse:

$$f_c = \frac{1}{T_r}$$

Generally:

$$f_c = \frac{k}{T_h}$$